Presentation 5

- Links to presentation(s) and code(s) on GitHub
  - Presentation PDF: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Dashboard/AggregateTrendsPresentation5_9Dec_Group1.pdf
  - Presentation video: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Aggregate%20Trends%20analysis/Aggregate_Trends_1_Presentation_5.mp4
  - Dashboard: https://app.powerbi.com/groups/me/reports/d9479cd4-f8f8-49a9-a794-7eb444e0bc1b/47de6a4fb2558b819092?experience=power-bi
  - Code: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Dashboard/MenuTraffic_%26_HourlyTrends_5Dec_EllaLiv.ipynb
- What did you do?
  - Cleaned more activity names in Python
  - Because there are different phone numbers to reach the LAC Main Menu and the Farmworker Menu, I combined the two menus in Python into one csv file
  - In Python, we determined the Contact Session IDs that never select any activity and added those to the data
  - Added Seniors Menu as a starting activity
  - Increased the number of activities to 22

    - Created the final dashboard to reflect all call journeys up to 22 activities
    - Looked through the dashboard to determine insights
        - There are many ways to reach Legal Menu 2, and only ~50% reach it the easiest way
        - Most callers who go to Legal Menu 1 then go to Legal Menu 2, implying that Legal Menu 1 is unnecessary
        - Seniors ADAPT is unnecessary
    - Helped create presentation to display insights
        - Created Recommendations slide
        - Added some insights to dashboard slides
- How does it help the project?
    - We determined our final recommendations for Cynthia for how to improve the Legal Aid calling system

Presentation 4

- Links to presentation(s) and code(s) on GitHub
    - Presentation PDF:

      https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Dashboard/AggregateTrendsPresentation4_19Nov_Group1.pdf
    - Presentation video:

      https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Aggregate%20Trends%20analysis/Aggregate_Trends_Presentation_4_Recording.mov
    - LAC Main Dashboard:

      https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Dashboard%20pbix%20files/LACMainDrill_20Nov_Ella.pbix
    - Pre-Legal Seniors Menu Dashboard:

      https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Dashboard%20pbix%20files/PreLegalSeniorsDrill_17Nov_Ella.pbix
    - Farmworker Menu Dashboard:

      https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Dashboard%20pbix%20files/FarmworkerDrill_17Nov_Ella.pbix
    - Code:

      https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Dashboard/MenuTraffic_%26_HourlyTrends_17Nov_Ella.ipynb

- What did you do?
    - Cleaned activity names in Python
        - There are several activity names such as "GetLoggedInSubSeniorOtherAgents" and "GetLoggedInOtherSubSeniorsAgents" – for these I combined both, so they are both now called "Sub Senior Other Queue"
        - Fixed Spanish activities so that they are now shown
    - Fixed the Pre-Legal Seniors dashboard to reflect the correct distinct count
    - Created the Farmworker Menu drill down with cleaned activity names
    - Created the LAC Main drill down with cleaned activity names and with a drill down to 8 activities
    - Helped come up with insights based on the drill downs
- How does it help the project?
    - We now have a drill down from the starting point, LAC Main, that goes down to 8 activities. This helps us see the overall process that callers go through from their starting point to their ending point.
- Issues faced (if any)
- Attempts to resolve issues (if any)
- Issues resolved (if any)
- Next steps
    - Continue analyzing trends to make more recommendations in the LAC Main dashboard drill down

- References (Mention if you built up on someone else’s work)

Presentation 3

- Links to presentation(s) and code(s) on GitHub
    - Presentation PDF:

      https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Dashboard/AggregateTrendsPresentation3_5Nov_Group1.pdf
    - Presentation video:

      https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Dashboard/Aggregate%20Trends%20Presentation%203.mp4
    - Pre-Legal Seniors Menu Dashboard:

      https://app.powerbi.com/groups/me/reports/fe9e0072-86e4-453c-9db5-75dd71219022/d01a1308fc4953bcd62a?experience=power-bi&clientSideAuth=0
    - Legal Menu Dashboard:

      https://app.powerbi.com/links/iwbseJmuNN?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&pbi_source=linkShare
    - Code:

      https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Code/Dashboard/MenuTraffic_%26_HourlyTrends_5Nov_Ella.ipynb
- What did you do?
    - Altered Krish's code to include fourth and fifth activities in our drill down menus
    - Added code for the Pre-Legal Seniors Menu to create a new cleaned dataset

    - Uploaded both new datasets to Power BI to create new drill down visuals for both the Legal Menu and the Pre-Legal Seniors Menu that show Activities 1-5
    - Came up with insights and recommendations based on the dashboards
        - Determined that Closed Queue is very common fairly quickly into the drill downs, so we should see if there is a way to navigate callers to other open agents
    - Helped create slides in the presentation showing the dashboard and showing the Next Steps and Further Questions
- How does it help the project?
    - We made progress creating drill downs and now understand how to alter the code to incorporate more EPs, which will make it easier to create drill downs for all menus
- Issues faced (if any)
- Attempts to resolve issues (if any)
- Issues resolved (if any)
- Next steps
    - Create drill downs for all other menus in the data, starting at Main Menu
- References (Mention if you built up on someone else's work)

Presentation 2

- Links to presentation(s) and code(s) on GitHub
    - Presentation: https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Dashboard/AggregateTrendsPresentation2_23Oct_Group1.pdf
    - Dashboard: https://app.powerbi.com/groups/me/reports/c3867d3c-9465-49f9-97a2-ca96b50050e8/91acb0b1e6532b002774?ctid=7d76d361-8277-4708-a477-64e8366cd1bc&experience=power-bi
- What did you do?
    - I helped with the data cleaning – we had to determine how to alter the dataset for the drill-down menu to work. We had to fill in the cells that said "N/A" with the appropriate EP name, Flow name, or Activity name. I realized that there is a separate row for each individual EP, Flow, and Activity, so we had to copy the EP and Flow names to the cells below in the data cleaning.
    - We created a drill down menu from EP name to flow name to activity name in Power BI.
    - I determined how the CAR dataset correlates to the Call Flows diagrams and created a slide demonstrating the correlation.
    - I thoroughly analyzed the Power BI dashboard to determine recommendations from our graphs. We determined that there are too many activity options in the Main Menu, adding a significant amount of time (58

seconds) to the call. The Main EP also has by far the most traffic, as all callers start there, so that should likely be the main focus.

    - I came up with some Next Steps and Further Questions, including grouping some activity names together so there are two separate groups for activities in the Main Menu, and clarifying with Cynthia what the Flow Names mean.

- How does it help the project?
    - This analysis helped us determine which areas of the call have the most and least traffic. This allows us to know what areas to focus on and which areas of the call get little enough traffic that they might be able to be combined.
- Issues faced (if any)
    - When there is a flow name that is not Unknown Flow, the EP name tends to be Unknown EP, and when the EP name is not Unknown EP, the flow name is almost always Unknown Flow. Because of this, we had trouble finding the patterns in the call flows initially.
- Attempts to resolve issues (if any)
- Issues resolved (if any)
    - In R, we cleaned the data to ensure that the correct EP, Flow, and Activity Name are in the same row.
- Next steps
    - Determine which activities under the Main Menu can be grouped together to make for more efficient call flows.
    - Group related menus together that receive little traffic.

- References (Mention if you built up on someone else’s work)

Presentation 1

- Links to presentation(s) and code(s) on GitHub
  - Presentation:
  - https://github.com/arvindkrishna87/STAT390_LegalAid_Fall2025/blob/main/Presentations/Dashboard/AggregateTrendsInitialAnalysis_7Oct_Group1.pdf
- What did you do?
  - I, along with the rest of my team, helped brainstorm new metric ideas for the project and which variables we would need to investigate for the proposed metrics. Specifically, I came up with the First-Route Success Rate metric, which measures how many calls are answered on the first attempt without being redirected. To calculate this, we will look at the column titled "Redirection Reason" and see all the calls that are "NA" and only have one unique Correlation ID, which I believe means the calls was not redirected, but we do need to clarify that. These calls would be "successes", and those that are redirected would be considered "failures". Redirection adds time to a call and can make for a worse experience for a caller, so this would be a valuable metric to look into.
  - I also helped come up with ideas for which variables to show on the graphs on the dashboard. For all four of the visualizations, I took part in coming up with the variables used and how to visualize them. We were interested in using the User Type column, and I came up with the idea of plotting it with Duration on the y-axis. I also decided we should not include the count of

    "NA" redirect reasons in the "Count of Redirection Reason" graph, as it made the other columns all appear small and more difficult to read. I created a list of Next Steps / Further Questions to determine where my team should look next to dive deeper into certain metrics or graphs we created, and I created the corresponding slide in the presentation.

- How does it help the project?
    - For this first presentation, we were trying to better understand the data in general. Through helping come up with graph ideas and new metrics, I looked deeply into the datasets and variables, and now I understand the variables much better and how they could be used to investigate the metrics. We now understand that for the calls that were redirected for the "Unconditional" or "Deflection" reasons, the call duration was much longer on average.
- Issues faced (if any)
    - We had trouble uploading the combined datasets into Power BI. There are also several variables with unclear explanations so we need to clarify those to fully understand what they demonstrate.
- Attempts to resolve issues (if any)
- Issues resolved (if any)
    - We initially used the most current All Calls (August 2025) and CAR (9/7/25-9/13/25) datasets, but once we were able to get access to the Power BI desktop version, this issue was resolved.
- Next steps

    - I will dive deeper into the First-Route Success Rate metric after calculating the number of successes and failures. For the failures and successes, I would further analyze other columns, such as Related Reason, User Type, and Call Outcome Reason to see if there are any patterns. For example, if a large portion of the failures are of the same User Type, then there might be an issue with the User Type column that we could help resolve.
    - We also still need to understand what those reasons and the “FollowMe” reason fully mean. We also want to ask if those calls with “NA” under Redirect Reason means that the calls were not redirected at all.
- References (Mention if you built up on someone else’s work)